Quiz — 1.3.4 Web Technologies — ANSWER SHEET

Separate answer sheet / mark scheme — keep for marking.

Answer key

Section A (8 marks, 1 each)

1. **B** — HTML defines structure and content.
2. **B** — CSS handles presentation/styling.
3. **C** — `#` selects by id (`.` selects by class, a bare name selects by tag).
4. **B** — JavaScript provides client-side interactivity/behaviour.
5. **B** — Each incoming link counts as a vote for importance.
6. **B** — Votes from high-rank pages carry more weight.
7. **C** — Verifying login/credentials against a database is server-side.
8. **B** — Client-side checks can be viewed/altered/bypassed, so are not secure on their own.

Section B (12 marks)

9. **[3] (1 mark each)** HTML: structure and content (headings, paragraphs, links, images). CSS: presentation/style (colour, font, layout). JavaScript: interactivity/behaviour, usually client-side (events, validation, dynamic updates).

10. **[2]** 1 mark: a web crawler/spider follows links and reads/visits pages. 1 mark: the engine builds an index mapping keywords to the pages they appear on for fast retrieval (and ranks by relevance).

11. **[2]** 1 mark: the damping factor (~0.85) models the probability a "random surfer" keeps following links rather than jumping to a random page. 1 mark: it prevents pages from gaining unrealistically high/looping rank and ensures the calculation converges to sensible values.

12. **[2]** 1 mark: the rank of each page is recalculated repeatedly using the latest rank values of the pages linking to it. 1 mark: it stops when the values stop changing significantly (converge).

13. **[2]** 1 mark advantage: faster/immediate response and reduced server load. 1 mark disadvantage: insecure (can be viewed/bypassed) / depends on the user's browser/device capabilities.

14. **[1]** Any valid: e.g. database queries, login authentication, or payment checks — because the data/logic must be kept secure and hidden, and cannot be trusted to the client.

Section C (5 marks — award up to 5 from)

15.
- The email format check is simple and not security-critical, so doing it client-side gives the user instant feedback. **[1]**
- Doing it client-side reduces load on the server because no round trip is needed for a simple check. **[1]**
- Payment verification must be done server-side because it is security-critical — client-side code can be viewed and bypassed by the user. **[1]**
- The bank's records/logic must stay hidden on the server, so the verification cannot be trusted to the browser. **[1]**
- Server-side processing is secure but slower (a round trip to the server) and adds server load — an acceptable trade-off for sensitive payment data. **[1]**

Total: 8 + 12 + 5 = 25 marks